

THIRTY THOUSAND YEARS

FOUR STRATA, ONE ENVIRONMENT
OWNER'S HANDBOOK

BUILD 260923.1
401 PARAMETERS

ON

DRONE

KEYS

MUTE

SOLO

MASS — WEIGHT

LEVEL
-2.9 dB

CUTOFF
364 Hz

RES
35 %

LADDER
FILTER

OSC 1
80 %

OSC 2
60 %

SUB
60 %

BEAT
0.356 Hz

DRIFT
10 c

ON

DRONE

KEYS

MUTE

SOLO

SIGNAL — ARTIFICIAL

LEVEL
-13.8 dB

CUTOFF
632 Hz

RES
30 %

TABLE 1
30 %

TABLE 2
20 %

PM INDEX
12 %

PARTIALS
0 %

SHIFT
0.0 Hz

INTERFERE
0 %

ON

DRONE

KEYS

MUTE

SOLO

MEMORY — TRACES

VOICE
SOURCE

GRAINS
MODE

LEVEL
-2.9 dB

POSITION
20 %

DENSITY
7.41 Hz

LENGTH
100 ms

JITTER
20 %

SCATTER
50 %

EROSION
0 %

ON

DRONE

KEYS

MUTE

SOLO

STRUCTURE — ARCHITECTURE

BEAM
BODY

MASS
EXCITER

LEVEL
-9.0 dB

DAMPING
45 %

STRESS
0 %

MATERIAL
70 %

COUPLING
20 %

DRIVE
0 %

STRIKE
70 %

STRIKE

HISTORY

TRAVELLING BY HAND - DRAG HISTORY, OR SET DRIVE TO AUTO TO LET IT RUN

AWAKENING
STORED
STORE RECALL CLEAR

OCCUPATION
STORED
STORE RECALL CLEAR

COLLAPSE
STORED
STORE RECALL CLEAR

AFTERMATH
STORED
STORE RECALL CLEAR

HISTORY
0 %

MANUAL
DRIVE

DURATION
20.05 s

8
BARS

ONCE
AT THE END

HISTORY
HOLD HIST
HOLD LIFE

SPECTRAL RECORD

48 BANDS · THE STRATA OF WHAT HAS BEEN PLAYED

An instrument named for how long spent fuel stays dangerous. Four ways of making sound, a shared environment that helps make it, a modulation network that listens to the engine's own output, and one control that carries the whole thing from one condition to another over up to half an hour. **September 2026.**

DISTANCE

LIFE

HUMANITY

WHAT IT IS

A PLACE, NOT A PATCH

Thirty Thousand Years makes sound four different ways at once, puts all four in the same room, and then lets the room take part in making the sound rather than only treating it.

The four ways are called **strata**. **MASS** is virtual analogue: two oscillators, a sub, and a beat set in hertz. It is the weight. **SIGNAL** is wavetables, phase modulation and a bank of additive partials — something has learned to think. **MEMORY** is granular and spectral reading over material you captured or imported — the world remembering itself. **STRUCTURE** is resonant bodies under an exciter — architecture under stress. Every one of them is a complete instrument, and any combination of them can be on at once.

Underneath them is the **ENVIRONMENT**: a mixer, two lanes of insert processors, a space, and an explicit **feedback loop** that can return into a stratum's input instead of into the mix. That last choice is the whole design in one control. A loop that returns into the mix is an echo. A loop that returns into STRUCTURE's exciter is the instrument playing itself.



The deck. The four strata across the top, HISTORY and the spectral record beneath them, the eight macros as one row, and the keyboard, the drone and the output along the foot.

INSTALLING

Copy **Thirty Thousand Years.vst3** into your VST3 folder — normally **C:\PROGRAM FILES\COMMON FILES\VST3** — and rescan. The standalone in the same download needs nothing installing.

THE FOUR STRATA



MAIN. One column per stratum, each with its own switches, its nine most-used controls and a live meter. Below them: HISTORY with its four scenes, and the spectral record — the output's own spectrum, drawn as layers of rock, left to right, oldest at the far edge.

MASS Two analogue oscillators with a sub and a beat measured in hertz. The weight the rest of the instrument is hung on, and the only stratum that is on when you open a fresh instance.

SIGNAL Two wavetables, a phase-modulation pair, a ring modulator and a 64-partial additive bank whose upper partials can lose their harmonic alignment while the fundamental holds.

MEMORY A granular reader and a spectral resynthesis over procedural material, a 30-second capture or an imported WAV, both under one EROSION control.

STRUCTURE Seven resonant bodies driven by an impulse, a noise burst, a bow, or any other stratum. STRESS carries it from a coherent object to fracture.

THE HEADER AND THE SIX VIEWS

The panel never changes shape. The header, the eight macros and the foot are always there; only the middle is swapped by the six tabs. Nothing is hidden in a menu, and every control on every view has a written explanation attached to it — press **TOOLTIPS** in the header to leave them on, or hold **CTRL** to see one while you point at something.



The header. The patch window with its name, category and note; then PATCHES, USER, SAVE and OPEN; the A/B pair with STORE; UNDO; MUTATE with its amount and its seven group locks; QUALITY; ASSIGN; MOTION; TOOLTIPS; WORLD FX; and PANIC.

MAIN	The four strata, HISTORY and the spectral record.
SOURCES	Four pages, one per stratum, with everything in it.
LIFE	Modulators, the network, the event lanes, the matrix.
HISTORY	The four scenes and the travel controls.
ENVIRONMENT	The insert lanes, the space and the feedback loop.
MIX	Four channel strips, output, tuning and voices.

PANIC

PANIC fades the output over 20 ms, clears every voice, every delay line and the feedback loop, switches the drone off, and stays stopped. Nothing restarts until you play a note or switch the drone back on. It is the one control in here that is allowed to be rude.

MUTATE, UNDO, A/B

MUTATE nudges the patch by the amount beside it. The seven chips lock a group out of it — MA, SI, ME, ST, EN, LI and GL for the globals, which is locked by default. Levels, the drone switch, the macros and the loop send are never mutated at all. **UNDO** swaps with the state before the last big move, so a second press is a redo.

THREE WAYS TO PLAY IT

A fresh instance is **exactly silent**, and stays silent until you ask it for something. That is deliberate: an instrument that starts droning the moment it is loaded cannot be used on a busy session. There are three ways to ask, and they share the same eight voices — the drone chord takes as many as its size needs and the keyboard gets the rest.



The foot. The keyboard with its octave buttons, held notes and voice lamps; the bend and mod wheels; DRONE with its root, chord, latch, spread and force; PLAY with the voice mode and glide; and OUTPUT with VOLUME, CEILING and the bass-mono fold.

DRONE

AN EXPLICIT SWITCH

Switch **DRONE** on and the instrument sounds with no note held. Set the **ROOT** as a note name and the **CHORD** above it, from UNISON to OPEN FIFTHS. **SPREAD** places those notes across the stereo field; **FORCE** is how hard they strike.

INSTRUMENT

UP TO EIGHT VOICES, MPE

POLY, MONO or LEGATO, with **velocity, bend, sustain and aftertouch**, and **MPE** for a controller that sends **per-note pressure and slide**. **VOICES** caps the count; **fewer voices costs less and makes a dense patch behave more like one object**.

EXCITATION

THE AUX INPUT

Feed audio into the AUX bus and set **AUX INPUT** on the MIX view. EXCITE drives STRUCTURE with it and feeds MEMORY's capture; DUCK makes the instrument step back for it; EXCITE + DUCK does both.

BOTH AT ONCE

Each stratum has its own **IN DRONE** and **ON KEYS** switches on the MIX view. That is how you build a patch where the drone is MASS and STRUCTURE while the keyboard plays SIGNAL over the top of it, out of one instance.

FIVE MINUTES

- ONE

In the header, step the patch window to **THE SEA IS MADE OF IRON**. Its note appears under the name.
- TWO

Switch **DRONE** on at the foot. It sounds. Nothing else has to be pressed.
- THREE

Take the **VIOLENCE** macro up slowly and back down. It is not a volume: the bowed plate is being pushed, not amplified.
- FOUR

Move **ROOT** down a few semitones and change **CHORD**. The whole world follows the root.
- FIVE

Open **SOURCES** > **STRUCTURE** and move **BOW SPEED**. You are now inside the instrument.

THEN TRY

THIRTY THOUSAND YEARS, the flagship. Its four scenes are already stored and **HISTORY** is already switched on. Set **DURATION** to thirty minutes, put **DRIVE** on **AUTO**, and leave it.



SOURCES. Everything a stratum owns, on one page, with nothing folded away.

EVERY CONTROL EXPLAINS ITSELF

This handbook describes what the instrument is for. It does not list all 401 parameters, because the panel already does: every control carries a written line saying what it is and how it is used, placed beside it and never over it. **TOOLTIPS** in the header leaves them showing; **CTRL** summons one on demand.

MASS — THE THING BENEATH THE GROUND



SOURCES › MASS. Two oscillators; WEIGHT and INSTABILITY; the filter with its circuit, drive and key tracking; and the two envelopes.

Two band-limited oscillators — sine, triangle, saw or pulse — each with an octave, a semitone, a fine tuning and a level, plus hard sync. Beneath them a sub one or two octaves down, and **REINFORCE**, which feeds MASS back into itself a little so that the low end survives a small speaker. The filter is either a four-pole **LADDER** or a state-variable circuit with low, band, high and notch, both with drive and both with saturation inside the resonance so a screaming filter stays a note.

BEAT (HZ)

Detunes oscillator 2 by a fixed number of hertz rather than a number of cents, 0.02 to 12 Hz. A drone at 55 Hz and a drone at 220 Hz then beat at exactly the same speed, which cents can never do.

DRIFT

Up to 50 cents of slow wander, with DRIFT TIME (0.2 to 60 s) setting how long it takes to get anywhere. Nothing here is meant to be perfectly stable.

SIGNAL — SOMETHING HAS LEARNED TO THINK



SOURCES › SIGNAL. Two wavetables; the phase-modulation pair and the 64-partial bank; DISPLACEMENT and the filter; the two envelopes.

Eight original wavetables — GEOMETRIC, MACHINE, FORMANT, HOLLOW, STEPPED, METALLIC, SIREN, ORGANISM — of eight frames each, on two oscillators that can ring-modulate one another. Behind them a two-operator phase-modulation pair with its own ratio, index and feedback, and an additive bank of up to 64 partials with controls for spread, odd against even, tilt, clustering, gaps and motion. **FUNDAMENTAL** is the one that matters: it holds the first partial in place while everything above it walks out of the harmonic series.

FREQ SHIFT

Moves every frequency by the same number of hertz, up to 2000 either way, and the knob is fine near zero. Harmonic relationships are destroyed, which is the point.

PITCH SHIFT

Transposes by a ratio instead, up to two octaves either way. These are two different operations and they have two different controls; do not reach for one expecting the other.

MEMORY — THE WORLD REMEMBERING ITSELF



SOURCES › MEMORY. The source and its capture buttons; the reader; the grain cloud; the spectral controls with EROSION under them.

Four procedural sources are built into the instrument — **VOICE**, **BROADCAST**, **CHOIR**, **MACHINE** — so a patch works with no files at all. Beyond them there is a rolling **CAPTURE** of the last thirty seconds, taken from before the loop, after the space, or from the AUX input, and an **IMPORT** of your own WAV. **MEMORY** reads that material with grains (5 ms to 2 s each, up to 200 a second), with a spectral resynthesis that can be frozen, smeared, tilted, thinned and displaced, or with both at once.

EROSION

One control over four kinds of decay — lost bandwidth, dropout, fragmentation and simplification — each of which stays separately reachable underneath. This is the stratum's whole argument on one knob.

REMEMBER

A capture is temporary until you press **REMEMBER**. Only then is it saved with the patch and the project. The panel says which state it is in.

STRUCTURE — ARCHITECTURE UNDER STRESS



SOURCES > STRUCTURE. The body and its material; the geometry and STRESS; the exciter with the bow controls; STRIKE, with a button that hits it now.

Seven bodies. **PLATE**, **CABLE**, **BEAM**, **CAVITY** and **GLASS** are banks of up to 48 tuned resonators; **TUBE** and **COMB** are waveguides. They are **designed** resonances rather than simulations of real objects, and they are meant to be pushed past where a real object would go. The exciter is an impulse, a noise burst, a bowing friction model, or any of MASS, SIGNAL, MEMORY, the loop return and the AUX input — which is where a stratum stops being a layer and becomes a cause.

STRESS

How much load the structure is under. Up, it detunes, then the modes start trading energy and it groans, then it fractures audibly.

SUSTAIN ENERGY

How much is fed in continuously rather than as a strike. It is what turns a struck body into one that keeps sounding, and it does not erase the strike.

THE SPACE, AND THE LOOP THAT GENERATES



ENVIRONMENT > SPACE & LOOP. The reverberator on the left, PERSPECTIVE in the middle, and the feedback loop on the right with its own energy meter.

The space is early reflections into an eight-line network, with a decay from a fifth of a second up to a minute, frequency-dependent damping, motion and a freeze. **APPARENT SCALE** and **DISTANCE** are separate on purpose: scale moves the reflection pattern, which is what the ear reads as the size of a place, while distance trades direct sound for reflected and loses the high end to the air. Neither of them is an amount, and neither touches the width.

WHERE THE LOOP RETURNS

RETURNS TO has four settings and it is the most consequential control in the environment. **MIX** makes the loop an echo. **STRUCTURE** hands it to the exciter, so the room plays the body. **MEMORY** feeds the capture, so the loop is remembered. **SIGNAL** feeds the ring modulator. The loop's own high-pass, low-pass, shift, saturation and damping decide whether it settles or climbs; the saturation is what lets it be run near unity without running away.

INSERT LANES



ENVIRONMENT > INSERT LANES. The two lanes on the left; every processor's controls are on the page beside them whether it is in a lane or not.

Two lanes of three slots. **Lane A** sits on the mix bus, **lane B** on the way into the space — so B is how you dirty the room without dirtying what stands in front of it. Any slot holds any of the seven, in any order, and an empty slot costs nothing.

SATURATE	Asymmetric valve saturation, level-matched by measurement so DRIVE is not simply louder.
FOLD	Wavefolding, symmetrical or not. It adds harmonics unrelated to the original spectrum.
MULTIBAND	Three bands driven separately, with the low band protected so the weight survives.
SHIFT	Single-sideband shifting in signed hertz, with feedback for a rising staircase.

DELAY	1 ms to 2 s. TAPE detunes and dulls every pass; CLEAN repeats what it was given.
COMB	A resonant comb tuned in semitones, with damping and diffusion.
CRUSH	Bit and rate reduction, with an anti-alias switch for the grit without the folding.

THE INSTRUMENT MOVES BY ITSELF



LIFE. Five pages: LFO, ENVELOPES & SHAPES, RANDOM & FOLLOW, EVENTS & NETWORK, and MATRIX.

8 LFOS	Six shapes, 0.0008 Hz to 50 Hz, so one cycle can last twenty minutes. Free or locked to the host's bars; global or one per voice.	4 RANDOM SOURCES	Smooth noise, a random walk with a restoring force, sample and hold, probability, cluster bursts, or bounded chaos.
4 ENVELOPES	Triggered by a note, by the drone starting, or by an event lane.	2 FOLLOWERS	Turn the loudness of any stratum, the mix or the AUX input into a modulation source.
4 SHAPES	Drawable multi-segment envelopes. Click to add a point, drag to move it, alt-click to remove it.	4 EVENT LANES	Each fires with a chance, at a rate, with a refractory wait, at one of seven targets.

THE MATRIX, AND THE NETWORK



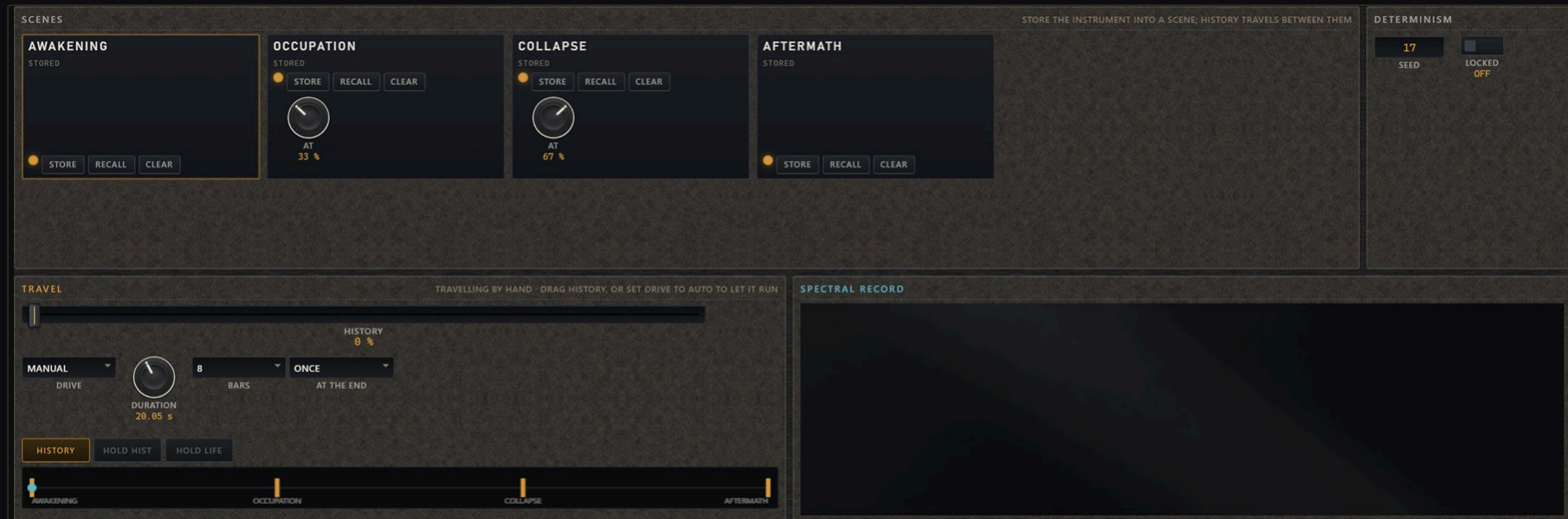
LIFE > MATRIX. Thirty-two slots. Each takes a source, a destination, a signed depth, a second source to scale it by, a curve, a slew and a range clamp.

Drag a source out of the palette on the RANDOM & FOLLOW page onto any control on any view and a slot is filled for you. Modulation is applied as an **offset** on the value you set, clamped to the parameter's own range — the value you edited never moves underneath you, so a modulated patch is still the patch you built.

THE LIFE NETWORK

Detectors measure energy, brightness and transients on each stratum and on the mix, and offer them as modulation sources like any other. **COUPLING** scales every one of those cross-stratum terms; **AUTONOMY** sets how far the world may drift from the state you edited; **RECOVERY** pulls it back. **HOLD LIFE** freezes every modulator where it stands while the sound goes on.

FOUR SCENES, ONE JOURNEY



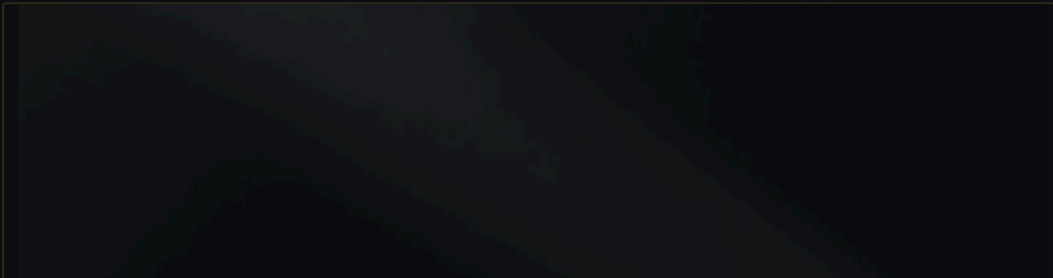
HISTORY. The four scenes along the top with their lamps and their names, the determinism controls beside them, TRAVEL below, and the spectral record to the right. The line under the scenes says what is stopping the journey, if anything is.

IT HAS TO BE ARMED, AND IT NEEDS TWO SCENES

HISTORY is off in a fresh patch and it needs at least two different scenes stored. Until both of those are true the HISTORY slider moves and nothing happens, which looks exactly like a broken control. The panel says so itself, in the line under the scene tiles: it will tell you there are no scenes stored, that there is only one, or that the scenes exist but HISTORY is switched off. Read that line first.

STORING SCENES, AND TRAVELLING

- ONE** Set the instrument up the way the piece should **start**. Press **STORE** on scene 1. Its lamp lights and the tile reads STORED.
- TWO** Change something — anything musical — and store scene 2. A scene holds every musical parameter; it deliberately does not hold the output calibration, the quality setting, the drone switch, the file state or the seeds.
- THREE** Switch **HISTORY** on. The slider now travels from scene 1 through the scenes you have stored. Scenes 2 and 3 can be moved along that line, so you can spend longer in one part of the piece than another.
- FOUR** Set **DRIVE**. **MANUAL** is you, by hand or by host automation. **AUTO** runs on its own clock over a **DURATION** of anything from one second to thirty minutes. **AUTO SYNC** runs on the host's bars instead, from 1 to 128 of them. At the end it can stop **ONCE**, **LOOP**, or **PING-PONG** back the way it came.



The spectral record. The output's own spectrum in forty-eight bands, laid down as strata — the oldest at the far edge. It is a picture of what has already happened.

HOW IT INTERPOLATES, AND HOW TO STOP IT

Pitches move in semitones, times and frequencies logarithmically, levels in decibels, and anything that is a list or a switch steps over at the midpoint rather than smearing. **HOLD HISTORY** stops the journey where it is without switching it off, so you can sit inside one moment for as long as you like.

EIGHT WORDS FOR WHAT HAPPENS NEXT



The macro row. Always visible, on every view. Click a macro's name to open its map and see or change what it reaches.

Each macro drives up to eight destinations at signed depths, and the map is stored in the patch. The meanings are consistent across the bank, so the same word does a recognisable thing on any preset — except **HUMANITY**, which is patch-specific by design and is mapped individually on the presets that use it.

MASS	Weight and physical presence: the low end, the sub and the body of everything.
DREAD	Suspense. The sound leans somewhere without arriving.
VIOLENCE	Abrasion and force — drive, stress and saturation. Level-matched, so it is not simply louder.
INSTABILITY	Loss of equilibrium: drift, wander, and the network's autonomy together.

CONTAMINATION	The strata stop being separate and start exciting each other.
DISTANCE	From near presence to a remote trace: space, damping, loss of detail.
LIFE	How much the instrument moves when you are not touching it.
HUMANITY	Recognisable vulnerability. What is left that a person could have made.

THE MIX, AND THE TWO FADERS



MIX. Four channel strips — level, pan, width, two cuts, the space and loop sends, and the participation switches — then OUTPUT, TUNING & VOICES, and anything the rest of the panel has no home for.

PATCH TRIM BELONGS TO THE PATCH; VOLUME BELONGS TO YOU

PATCH TRIM on the MIX view is the patch's own output level. It is saved with the patch, it travels through HISTORY, and it is what the forty-eight factory patches are levelled with, so one of them is not three times louder than the next. **VOLUME** at the foot is yours, and no preset ever touches it. **VOLUME** runs **into** the limiter, so pushing it past the point where the meter stops rising is asking for compression rather than for level. The limiter is idle on every factory patch, and the whole bank sits inside a 16.8 dB spread.

THE BROKILD WORLD FX RACK

WORLD FX in the header opens the rack that every Brokild instrument shares. It sits after the engine's own environment, at the very end of the chain, and it is **empty by default** — a patch that does not use it sounds exactly as though it were not there.

The rack holds reorderable effect pedals on the left and SPECTRA characters on the right, each with its own dry/wet PRESENCE, and it has its own presets. It is stored inside the patch, so a patch you save carries its rack with it.

FIVE AUTOMATABLE MACROS

BWFX MACRO 1 to 5 are the rack's entire host-parameter surface, and they bring the total offered to a host to 401. Each one maps a destination from one end of its range to the other, so a macro at 100 % runs its control all the way up and a macro at 50 % stops it halfway. Macro 5 ships wired to the rack's own dry/wet, so a fresh rack is exactly today's sound.

WHERE IT SITS	After the mix, the space, the loop and the limiter. It is the last thing that happens.
WHAT IT COSTS	Nothing, until a module is switched on.
REORDERING	Drag a pedal by its grip, or use its UP and DN buttons. Order is part of the sound.
SPECTRA	Characters that reshape how every voice is built rather than sitting in the chain. MASS and SIGNAL consume their modulation bus.

IT IS NOT THE ENVIRONMENT

The rack is a finishing chain. The insert lanes, the space and the feedback loop are part of how the instrument makes its sound, and they are not in here. If you want the room to generate, use the ENVIRONMENT view.

FORTY-EIGHT PATCHES, ONE

DRONE		INDUSTRIAL PULSE	
Beneath Thirty Kilometres of Concrete	Sub pressure under immense resonances.	An Intelligence in the Walls	Pulses that learn to answer each other.
The Sea Is Made of Iron	A bowed plate over a sub; the bow breathes.	Something Learned to Breathe	Adapting pulses out of noise.
Orbital Debris Choir	Inharmonic tones; the fundamental holds.	A Weapon Waiting for a War	Contained energy, then a release.
Steel Cathedral	A plate the size of a building, bowed.	Hostile Computation	A machine thinking out loud.
Abandoned Substation	Mains hum, a transformer, a warm loop.	Breath of the Grid	A synced gate over a sub. Tempo-locked.
Hollow Earth	Hollow tables in a cavity. A mountain's note.	Telemetry	Stepped tables, crushed, a siren behind.
Cooling Towers	Concrete cylinders breathing, wide and slow.	The Last Generator	A gate, a tape delay, the loop in the body.
Monolith	One enormous square sub. Nothing moves.	RESTRAINED BED	
EVOLVING WORLD		A City Without Witnesses	Distant structures and air.
The Machines Kept Working	A process that organises itself.	Population: Zero	Almost silence. Turn it up.
No One Answers the Broadcast	A message degrades into fragments.	The Sun Behind the Ash	Obscured light. Nothing is loud.
The Factory Dreamed of Flesh	Machine resonances acquire formants.	Granite Pad	A dry wide pad. Sits under drums.
Thirty Thousand Years	The flagship journey. Thirty minutes.	Voices Under the Ice	Frozen, smeared voices under a sine.
Furnace	Heat rising through a beam.	Last Light in the Tower	One fragile sine on glass modes.
Swarm	Organism tables; the network wide awake.	The Deep Field	Inharmonic partials rising very slowly.

FORTY-EIGHT PATCHES, TWO

TRANSITIONAL TEXTURE

The Last Human Frequency	A formant survives hostile spectra.
We Were Promised a Future	A mournful chord goes into the machine.
Sirens Over the District	Sirens sweeping, far away.
The Archive Burns	A frozen choir while the crusher eats it.
Slow Collapse	A clean chord to rubble in four minutes.
Radio Silence	The broadcast, frozen and shifted, thins out.

RESONANT GESTURE

After the Final Transmission	Strike once; the loop takes the residue.
The Room Is Listening	Every note strikes glass. Play sparsely.
Stressed Plate	Strike it and it fractures.
Glass Rain	Noise bursts on glass, scattered by a lane.
Plate Tectonics	Bowed so slowly it creaks, until it gives.

PLAYABLE BASS

Concrete Bass	Dry sub bass with a ladder growl. Mono.
Iron Bass	A struck cable. Velocity is the hammer.
Cable Bass	A bowed cable; aftertouch presses the bow.
Servo Bass	A machine table, snapping filter, a comb.

DRY

Cold Start	The naked MASS oscillators. A start.
Naked Signal	Only the SIGNAL tables, clean. For building.
Naked Structure	A struck plate and nothing else.
Naked Memory	The voice source through grains, dry.
Thin Air	A sine and a triangle, slowly. The quiet tone.

TEN ARE ALREADY TRAVELLING

Ten of the forty-eight ship with their four scenes stored and HISTORY already switched on. On those, set a DURATION and put DRIVE on AUTO and the piece runs itself. The rest wait for you to store scenes of your own.

SPECIFICATIONS

Formats	VST3 and Standalone, Windows 64-bit
Build	260923.1
Host parameters	401 – 396 plus five BwFX macros
Factory patches	48, ten with scenes stored
Voices	1 to 8, POLY / MONO / LEGATO, MPE
Audio in	One stereo AUX bus
Latency	88 samples at 44.1 kHz (2 ms of lookahead)
Sample rates	44.1 / 48 / 88.2 / 96 kHz
Cost, one core	5.6 % for a two-voice drone; 22.1 % for the flagship at eight voices with every macro up
Verification	67 engine checks, 90 panel checks, 16 host checks, all clear

REQUIREMENTS

Windows 10 or 11, 64-bit, and a VST3 host. The panel is drawn by the **Microsoft WebView2** runtime, which is present on any up-to-date Windows 10 or 11 installation; if it is missing the plugin says so and tells you where to get it. The engine does not depend on the panel: a patch runs identically with the editor shut.

CREDITS

Designed and built by Brokild, September 2026, with the Brokild World FX rack. Named for how long spent nuclear fuel stays dangerous. Written under the long shadow of Asimov, Gibson, Bradbury and Huxley, and in the weight of Cult of Luna and LLNN.

BROKILD · THIRTY THOUSAND YEARS · 260923.1